Arrays and Structures

Learning Objectives

After learning this chapter, the students will be able to

- Know the structured data type using arrays.
- Know the types of arrays.
- Writing programs to manuplates different types of arrays.

12.1 Introduction

The variables are used to store data. These variables are the one of the basic building blocks in C++. A single variable is used to store a single value that can be used anywhere in the memory. In some situations, we need to store multiple values of the same type. In that case, it needs multiple variables of the same data type. All the values are stored randomly anywhere in the memory.

For example, to store the roll numbers of the 100 students, it needs 100 variables named as roll1, roll2, roll3,……. roll100 . It becomes very difficult to declare 100 variables and store all the roll numbers. In C++, the concept of Array helps to store multiple values in a single variable. Literally, the meaning of **Array is "More than one". In other words, array is an easy way of storing multiple values of the same type referenced by a common name". An array is also a derived data type in C++.**

"An array is a collection of variables of the same type that are referenced by a common name". In an array, the values are stored in a fixed number of elements of the same type sequentially in memory. Therefore, an integer array holds a sequence of integers; a character array holds a sequence of characters and so on. The size of the array is referred to as its dimension.

12.2 Types of Arrays:

There are different types of arrays used in C++. They are:

- One-dimensional arrays
- Two-dimensional arrays
- Multi-dimensional arrays

12.2.1 One-dimensional array

This is the simplest form of an array. A one dimensional array represents values that are stored in a single row or in a single column.

Declaration

Syntax:

<data type><array_name> [<array_size>];

data_type declares the basic type of the array, which is the type of each element in the array.

array_name specifies the name with which the array will be referenced.

array_size defines how many elements the array will hold. Size should be specified with square brackets [ ].

Example:

```
int num[10];
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